

Simulated default risk and review capacity

Estimate a default event within 90 days

Application-time features support a simulated review queue.
A 12% batch ceiling limits the number of review flags.

Final result: Logistic Regression
2,500 challenge scores and 300 simulated review flags

Synthetic data only. A flag does not approve or decline financing.

Forward validation and reserved calibration

6,576 fit and selection rows

836 reserved calibration rows, including 78 positives

Three outer validation periods: 2023Q1, 2023Q3, 2024Q1
2,155 common outer OOF rows compare six candidates.

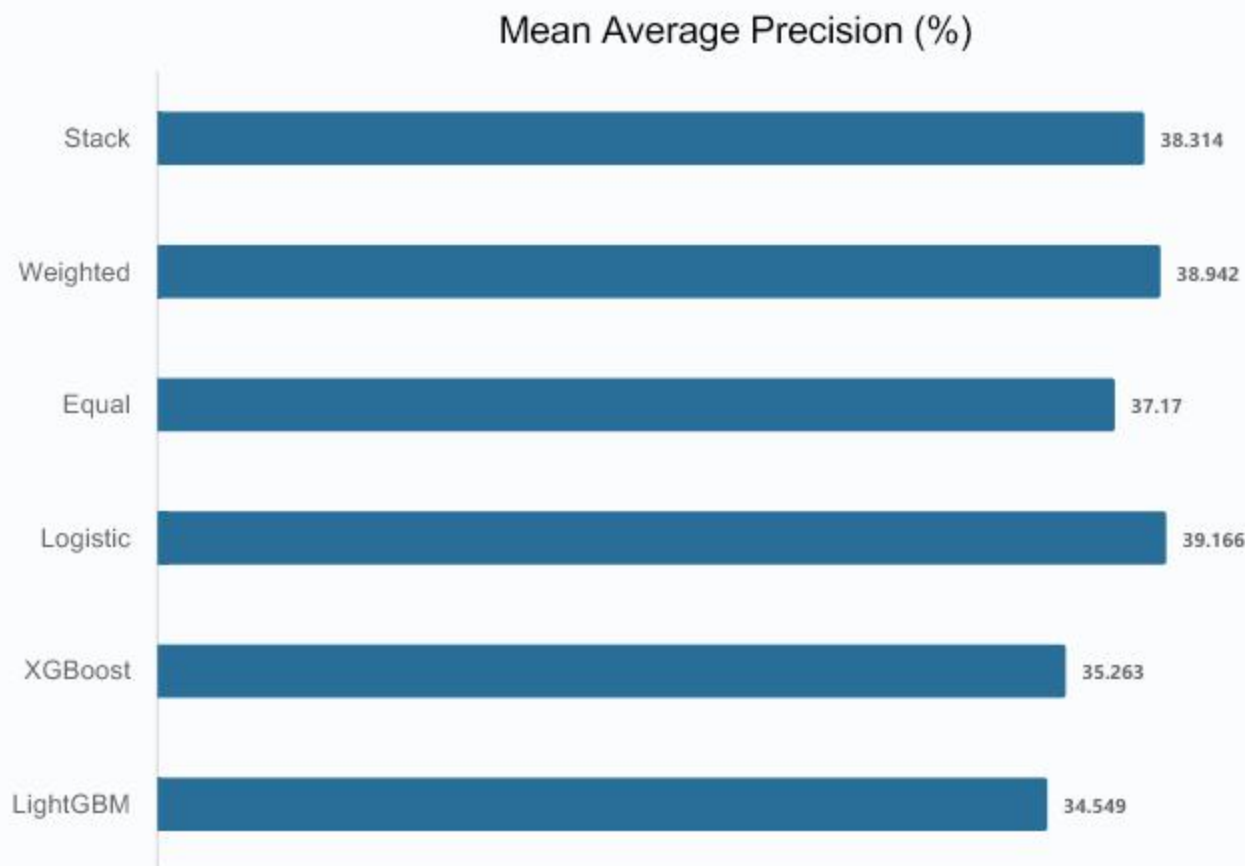
Training uses earlier, mature 90-day outcomes.

Customer separation applies to outer and inner splits.

Weights and the stacker learn only from inner OOF predictions.

OOF supports development selection. Earlier course work used these data.

Logistic outperformed the ensemble procedures



KEEP SINGLE

Logistic AP: 0.39166

Fold SD: 0.02981

No ensemble passed the gate.

OOF policy: 245 flags

84 TP, 161 FP, 95 FN

Loss: $10 \times 95 + 161 = 1,111$

Highest period: 11.749%

Fold SD is descriptive. It is not a confidence interval or significance test.

Calibration and explanation have limited scope

Final Logistic calibration-fit diagnostics

Brier: 0.076473 raw, 0.078058 after sigmoid
ECE: 0.021121 raw, 0.034871 after sigmoid
Both worsened. These rows also fitted the calibrator.

Day 4 case TR-009585

Bureau score 497 and DTI 1.2806 raised the earlier model score.
Raw probability 0.9031, calibrated probability 0.4795.
These SHAP reasons do not explain the final Logistic model.

Frozen batch policy and final limitations

330 threshold candidates, 300 final flags

Calibrated threshold: 0.12225843144286948

One full-batch cap removed 30 candidates.

Equal-score blocks stay together, without ID tie-breaking.

Regional OOF FPR gap: 3.976 percentage points

Descriptive evidence without intervals or fairness certification.

Challenge labels are unavailable, so challenge AP and loss are unknown.

Monitor missingness, score drift and queue load. After labels mature, review calibration and regional errors before a new validated version.